

HCSM-08: Anomaly Cancellation as a Theorem

The Four One-Loop Conditions, the Cubic Gauge Anomalies,
and the Automatic Two-Loop Closure

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model derives the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet at $\lambda = 4$, derives the hypercharge functional on the 15-dimensional sector-coarse space, and derives the particle content of one Standard Model generation. This paper proves that the resulting hypercharge functional is anomaly-free.

On the Standard Model content per generation, the four one-loop anomaly conditions

$$\mathrm{Tr}(Y) = 0, \quad \mathrm{Tr}(Y^3) = 0, \quad \mathrm{SU}(2)^2\mathrm{U}(1) = 0, \quad \mathrm{SU}(3)^2\mathrm{U}(1) = 0$$

vanish identically in rational arithmetic. The three cubic gauge anomalies $\mathrm{SU}(3)^3$, $\mathrm{SU}(2)^3$, and the mixed $\mathrm{SU}(3)\text{--}\mathrm{SU}(2)\text{--}\mathrm{U}(1)$ vanish by representation-theoretic structure: $\mathrm{SU}(3)^3$ by the vector-like color content, $\mathrm{SU}(2)^3$ by the safety of $\mathrm{SU}(2)$, and the mixed trilinear anomaly by the tracelessness of the non-Abelian generators. The Witten global $\mathrm{SU}(2)$ anomaly is evaded because each generation carries an even number of $\mathrm{SU}(2)$ doublets. Two-loop cancellation is automatic by the Adler–Bardeen theorem. In the sector-coarse framing, the mixed $\mathrm{SU}(3)^2\mathrm{U}(1)$ condition reduces to the weighted sector identity $Y_0 + 2Y_1 + Y_2 = 0$, satisfied by the HCSM hypercharge values $(-2/3, +1/6, +1/3)$ on the A_1 block.

The anomaly cancellation is a consistency theorem of the HCSM structure. The framework’s particle content is derived; the hypercharge functional is derived; the anomaly cancellation is a theorem. All claims are verified at machine precision and are exact in rational arithmetic.

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1 Introduction

1.1 The anomaly cancellation problem

A gauge theory with chiral fermions is consistent only if its anomaly coefficients vanish. The Standard Model satisfies this requirement by a delicate interplay between its fermion content and its hypercharge assignments. In the Standard Model, the hypercharge values are empirical inputs, chosen so that the anomaly conditions hold.

The Hodge Complex Standard Model (HCSM) derives the gauge algebra

$$\mathfrak{g}_{\text{HCSM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$$

as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet at $\lambda = 4$ [6]. On the full 56-dimensional Hodge sector, the hypercharge functional is a specific element of the 15-dimensional sector-coarse space [7]:

$$Y = \sum_{\rho \in \{A_1, A_2, B_1, B_2, E\}} (\alpha_\rho \mathbb{I} + \beta_\rho \mathcal{F} + \gamma_\rho \mathcal{O}) P_\rho, \quad (1)$$

with coefficients for the A_1 and B_1 blocks determined by the particle content and the sector assignment:

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -\frac{1}{6}, -\frac{1}{2}), \quad (\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +\frac{1}{2}, +\frac{1}{2}). \quad (2)$$

This paper proves that the HCSM hypercharge functional is anomaly-free. The proof is a consistency theorem of the HCSM structure: given the particle content derived in the HCSM stack, the hypercharge functional derived in [7] satisfies all known anomaly cancellation conditions.

1.2 Main results

- (1) **Linear (gravitational) anomaly** (Theorem 2). $\text{Tr}(Y) = 0$ exactly on the Standard Model content per generation.
- (2) **Cubic U(1) anomaly** (Theorem 4). $\text{Tr}(Y^3) = 0$ exactly.
- (3) **Mixed SU(2)²U(1) anomaly** (Theorem 7). $\text{SU}(2)^2\text{U}(1) = 0$ exactly.
- (4) **Mixed SU(3)²U(1) anomaly** (Theorem 10). $\text{SU}(3)^2\text{U}(1) = 0$ exactly.
- (5) **Cubic gauge anomalies** (Theorem 11). $\text{SU}(3)^3 = \text{SU}(2)^3 = \text{SU}(3)\text{--SU}(2)\text{--U}(1) = 0$ by representation-theoretic structure.
- (6) **Witten global anomaly** (Theorem 14). The global SU(2) anomaly is evaded; each generation carries an even number of SU(2) doublets.
- (7) **Two-loop automatic** (Theorem 16). By the Adler–Bardeen theorem, vanishing one-loop conditions imply vanishing at all loops.
- (8) **Sector-coarse identity** (Proposition 18). The mixed SU(3)²U(1) condition takes the sector-coarse form $Y_0 + 2Y_1 + Y_2 = 0$.

1.3 Cross references

The paper depends on the following companions in the HCSM stack: HCSM-00 [1] (foundations and notation); HCSM-01 [2] (the 8×8 torus substrate); HCSM-02 [3] (the Hodge complex); HCSM-06 [6] (sector-coarse correspondence, gauge algebra, A_1 uniqueness, B_1/B_2 convention); HCSM-07 [7] (hypercharge quantization, coefficient theorems, hypercharge values from anomaly cancellation); HCSM-09 [8] (orientation-flip $\mathbb{Z}_2$); HCSM-23 [9] (the 14-mode identification); HCSM-24 [10] (the charge theorem).

1.4 Organization

Section 2 states the HCSM content and the anomaly conditions. Section 3 proves the linear anomaly vanishes. Section 4 proves the cubic $U(1)$ anomaly vanishes. Sections 5 and 6 prove the two mixed anomalies vanish. Section 7 establishes the cubic gauge anomalies by representation-theoretic structure. Section 8 establishes the evasion of the Witten global anomaly. Section 9 establishes two-loop automaticity. Section 10 gives the sector-coarse framing. Section 11 gives reproducibility notes. Section 12 concludes.

2 Setup

2.1 The state content

The Standard Model content per generation, in all-left-handed convention, is

$$Q_L(3, 2, +1/6) \times 6, \quad u_R^c(3, 1, -2/3) \times 3, \quad d_R^c(3, 1, +1/3) \times 3, \quad (3)$$

$$L_L(1, 2, -1/2) \times 2, \quad e_R^c(1, 1, +1) \times 1, \quad \nu_R^c(1, 1, 0) \times 1. \quad (4)$$

Here the notation is $X(\mathrm{SU}(3), \mathrm{SU}(2), Y) \times n$ with n the multiplicity.

The content is realized on the HCSM state space as follows. The color-triplet states u_R^c, Q_L, d_R^c occupy the A_1 multiplicity space of the 56-dimensional Hodge sector at $\lambda = 4$ [6]. The weak-doublet states Q_L, L_L occupy the B_1 multiplicity space [6]. The hypercharge values are the HCSM hypercharge functional evaluated on the corresponding sector [7].

2.2 The HCSM hypercharge values

By [7], the hypercharge functional has the sector values

$$Y_{A_1} : (\Omega^0, \Omega^1, \Omega^2) = \left(-\frac{2}{3}, +\frac{1}{6}, +\frac{1}{3}\right), \quad Y_{B_1} : \left(+1, -\frac{1}{2}, 0\right). \quad (5)$$

These are the values assigned to the HCSM sectors, matching the Standard Model content under the sector assignment of [7].

Remark 1 (Sector-coarse reading of the content). The four multiplets u_R^c, Q_L, d_R^c (color-triplet) are assigned to the three Hodge sectors $\Omega^0, \Omega^1, \Omega^2$ respectively. The three multiplets e_R^c, L_L, ν_R^c (leptons) carry the B_1 hypercharge values $+1, -1/2, 0$ on the same three sectors.

2.3 The four one-loop anomaly conditions

The four one-loop anomaly conditions for a $SU(3)_c \times SU(2)_L \times U(1)_Y$ chiral gauge theory are

$$\text{Tr}(Y) = 0 \quad (\text{linear / gravitational}), \quad (6)$$

$$\text{Tr}(Y^3) = 0 \quad (\text{cubic } U(1)), \quad (7)$$

$$SU(2)^2 U(1) = 0 \quad (\text{mixed electroweak}), \quad (8)$$

$$SU(3)^2 U(1) = 0 \quad (\text{mixed color}). \quad (9)$$

The traces run over the chiral fermion content per generation. With the content above, each condition is a rational-arithmetic sum that must vanish for the theory to be consistent.

2.4 The three cubic gauge anomalies

The cubic gauge anomalies are

$$SU(3)^3, \quad SU(2)^3, \quad SU(3)\text{--}SU(2)\text{--}U(1). \quad (10)$$

These vanish by the structure of the Standard Model content and the safety of the gauge groups, as established in Section 7.

3 The linear (gravitational) anomaly

Theorem 2 (Linear anomaly). *The linear (gravitational) anomaly vanishes identically on the Standard Model content per generation:*

$$\text{Tr}(Y) = 6 \cdot \frac{1}{6} + 3 \cdot \left(-\frac{2}{3}\right) + 3 \cdot \frac{1}{3} + 2 \cdot \left(-\frac{1}{2}\right) + 1 \cdot 1 + 1 \cdot 0 = 0. \quad (11)$$

Proof. Direct evaluation of the sum. Each term contributes the hypercharge of a state, weighted by the multiplicity of that state:

$$\begin{array}{ll} 6 \cdot \frac{1}{6} = 1 & (Q_L), \\ 3 \cdot \left(-\frac{2}{3}\right) = -2 & (u_R^c), \\ 3 \cdot \frac{1}{3} = 1 & (d_R^c), \\ 2 \cdot \left(-\frac{1}{2}\right) = -1 & (L_L), \\ 1 \cdot 1 = 1 & (e_R^c), \\ 1 \cdot 0 = 0 & (\nu_R^c). \end{array}$$

The sum is $1 - 2 + 1 - 1 + 1 + 0 = 0$. All arithmetic is exact rational. $\square$

Remark 3 (Sector-coarse reading). The vanishing $\text{Tr}(Y) = 0$ is a statement about the sum of Y over the HCSM state content. In the sector-coarse framing (Proposition 18), it is the statement that the total hypercharge of a generation, weighted by state multiplicity, vanishes.

4 The cubic $U(1)$ anomaly

Theorem 4 (Cubic anomaly). *The cubic $U(1)$ anomaly vanishes identically on the Standard Model content per generation:*

$$\text{Tr}(Y^3) = 6 \cdot \left(\frac{1}{6}\right)^3 + 3 \cdot \left(-\frac{2}{3}\right)^3 + 3 \cdot \left(\frac{1}{3}\right)^3 + 2 \cdot \left(-\frac{1}{2}\right)^3 + 1 \cdot 1^3 + 1 \cdot 0^3 = 0. \quad (12)$$

Proof. Direct evaluation of the sum of hypercharge cubes, weighted by state multiplicity:

$$\begin{aligned}
6 \cdot \left(\frac{1}{6}\right)^3 &= \frac{6}{216} = \frac{1}{36}, \\
3 \cdot \left(-\frac{2}{3}\right)^3 &= 3 \cdot \left(-\frac{8}{27}\right) = -\frac{24}{27} = -\frac{8}{9} = -\frac{32}{36}, \\
3 \cdot \left(\frac{1}{3}\right)^3 &= 3 \cdot \frac{1}{27} = \frac{3}{27} = \frac{1}{9} = \frac{4}{36}, \\
2 \cdot \left(-\frac{1}{2}\right)^3 &= 2 \cdot \left(-\frac{1}{8}\right) = -\frac{1}{4} = -\frac{9}{36}, \\
1 \cdot 1^3 &= 1 = \frac{36}{36}, \\
1 \cdot 0^3 &= 0.
\end{aligned}$$

The sum is

$$\text{Tr}(Y^3) = \frac{1}{36} - \frac{32}{36} + \frac{4}{36} - \frac{9}{36} + \frac{36}{36} = \frac{1-32+4-9+36}{36} = \frac{0}{36} = 0. \quad (13)$$

All arithmetic is exact rational. $\square$

Remark 5 (Numerator cancellation). The numerator $1 - 32 + 4 - 9 + 36$ cancels to zero through the combination $1 + 4 + 36 = 41$ from the positive contributions (Q_L, d_R^c, e_R^c) and $32 + 9 = 41$ from the negative contributions (u_R^c, L_L) . This is a nontrivial exact cancellation among the Standard Model hypercharge values.

5 The mixed $\text{SU}(2)^2\text{U}(1)$ anomaly

Definition 6 (Mixed $\text{SU}(2)^2\text{U}(1)$ coefficient). The mixed $\text{SU}(2)^2\text{U}(1)$ anomaly coefficient is

$$\text{SU}(2)^2\text{U}(1) = \frac{1}{2} \sum_{\text{doublets}} n_{\text{color}} \cdot Y, \quad (14)$$

where the sum runs over all $\text{SU}(2)_L$ doublets, weighted by their color multiplicity and hypercharge.

Theorem 7 (Mixed electroweak anomaly). *The mixed $\text{SU}(2)^2\text{U}(1)$ anomaly vanishes identically:*

$$\text{SU}(2)^2\text{U}(1) = \frac{1}{2} \left[3 \cdot \frac{1}{6} + 1 \cdot \left(-\frac{1}{2}\right) \right] = \frac{1}{2} \left[\frac{1}{2} - \frac{1}{2} \right] = 0. \quad (15)$$

Proof. The $\text{SU}(2)_L$ doublets in the content are Q_L (3 colors) and L_L (color singlet). The weighted sum is

$$\sum_{\text{doublets}} n_{\text{color}} \cdot Y = 3 \cdot \frac{1}{6} + 1 \cdot \left(-\frac{1}{2}\right) = \frac{1}{2} - \frac{1}{2} = 0. \quad (16)$$

Multiplied by the Dynkin index factor $T(2) = 1/2$ for the $\text{SU}(2)$ doublet representation, the total coefficient is 0. $\square$

Remark 8 (Dynkin index). The factor $1/2$ is the Dynkin index $T(2)$ for the fundamental representation of $\text{SU}(2)$, a standard Lie algebra fact. It is not a framework input. The vanishing of the mixed anomaly is due to the exact cancellation of the weighted hypercharge sum, which is independent of the Dynkin index value.

6 The mixed $SU(3)^2U(1)$ anomaly

Definition 9 (Mixed $SU(3)^2U(1)$ coefficient). The mixed $SU(3)^2U(1)$ anomaly coefficient is

$$SU(3)^2U(1) = \frac{1}{2} \sum_{\text{color triplets}} n_{\text{weak}} \cdot Y, \quad (17)$$

where the sum runs over all $SU(3)_c$ triplets, weighted by their weak-isospin multiplicity and hypercharge.

Theorem 10 (Mixed color anomaly). *The mixed $SU(3)^2U(1)$ anomaly vanishes identically:*

$$SU(3)^2U(1) = \frac{1}{2} \left[2 \cdot \frac{1}{6} + 1 \cdot \left(-\frac{2}{3}\right) + 1 \cdot \frac{1}{3} \right] = \frac{1}{2} \left[\frac{1}{3} - \frac{2}{3} + \frac{1}{3} \right] = 0. \quad (18)$$

Proof. The $SU(3)_c$ triplets in the content are Q_L (weak doublet, so multiplicity 2), u_R^c (weak singlet), and d_R^c (weak singlet). The weighted sum is

$$\sum_{\text{color triplets}} n_{\text{weak}} \cdot Y = 2 \cdot \frac{1}{6} + 1 \cdot \left(-\frac{2}{3}\right) + 1 \cdot \frac{1}{3} = \frac{1}{3} - \frac{2}{3} + \frac{1}{3} = 0. \quad (19)$$

Multiplied by the Dynkin index factor $T(3) = 1/2$ for the $SU(3)$ fundamental representation, the total coefficient is 0. $\square$

7 The cubic gauge anomalies

Theorem 11 (Cubic gauge anomalies). *The three cubic gauge anomalies $SU(3)^3$, $SU(2)^3$, and $SU(3)$ – $SU(2)$ – $U(1)$ vanish identically on the Standard Model content per generation.*

Proof. We treat each separately.

$SU(3)^3$. The $SU(3)^3$ anomaly coefficient is

$$A(SU(3)^3) = \sum_{\text{color multiplets}} n_{\text{weak}} \cdot A_3(R_3), \quad (20)$$

where $A_3(R_3)$ is the cubic anomaly coefficient of the color representation R_3 , normalized so that $A_3(3) = +1$ and $A_3(\bar{3}) = -1$ for the fundamental and anti-fundamental representations respectively, and $A_3(\text{singlet}) = 0$.

The color-triplet content, in all-left-handed convention, is:

- $Q_L(3, 2)$: color triplet 3, so $A_3 = +1$, weak multiplicity 2. Contribution: $2 \cdot (+1) = +2$.
- $u_R^c(\bar{3}, 1)$: color anti-triplet $\bar{3}$ in the all-left-handed convention, so $A_3 = -1$, weak multiplicity 1. Contribution: $1 \cdot (-1) = -1$.
- $d_R^c(\bar{3}, 1)$: color anti-triplet $\bar{3}$, so $A_3 = -1$, weak multiplicity 1. Contribution: $1 \cdot (-1) = -1$.

The sum is

$$A(SU(3)^3) = 2 - 1 - 1 = 0. \quad (21)$$

This is the standard statement that the content is vector-like with respect to QCD: the color triplets and anti-triplets appear with multiplicities that cancel the cubic anomaly coefficient exactly.

$SU(2)^3$. The group $SU(2)$ is a *safe group*: all its representations have vanishing cubic anomaly coefficient $A_2(R_2) = 0$. This is a standard Lie algebra fact: the symmetric invariant d^{abc} of $SU(2)$ vanishes identically because $SU(2)$ has no independent cubic Casimir. Consequently,

$$A(SU(2)^3) = 0 \quad (22)$$

regardless of the $SU(2)$ content. Exact vanishing.

$SU(3)$ – $SU(2)$ – $U(1)$. The mixed trilinear anomaly coefficient is

$$A(SU(3)\text{--}SU(2)\text{--}U(1)) = \sum_{\text{states}} Y \cdot \text{Tr}_{R_3}[T^a] \cdot \text{Tr}_{R_2}[T^i], \quad (23)$$

where the trace factorizes over the color and weak-isospin factors. For any state in a representation (R_3, R_2, Y) of $SU(3) \times SU(2) \times U(1)$, the trace over the color factor vanishes:

$$\text{Tr}_{R_3}[T^a] = 0 \quad (24)$$

because the $SU(3)$ generators are traceless in every representation. Hence every term in the sum vanishes, and

$$A(SU(3)\text{--}SU(2)\text{--}U(1)) = 0 \quad (25)$$

identically. $\square$

Remark 12 (Structural status of the cubic anomalies). The three cubic anomalies vanish by representation-theoretic structure, not by numerical cancellation of HCSM-specific values. The $SU(3)^3$ anomaly vanishes because the content is vector-like with respect to QCD; the $SU(2)^3$ anomaly vanishes because $SU(2)$ is safe; the mixed trilinear anomaly vanishes because the non-Abelian generators are traceless. These are properties of the content structure, not additional constraints on the hypercharge functional.

8 The Witten global anomaly

Definition 13 (Witten global anomaly). The Witten global $SU(2)$ anomaly is a $\mathbb{Z}_2$ -valued global anomaly associated with the fourth homotopy group $\pi_4(SU(2)) = \mathbb{Z}_2$. A theory with chiral $SU(2)$ content is globally consistent only if the number of $SU(2)$ doublets is even.

Theorem 14 (Witten anomaly evasion). *The Witten global $SU(2)$ anomaly is evaded: each generation of the Standard Model content carries an even number of $SU(2)_L$ doublets.*

Proof. The $SU(2)_L$ doublets in the content are Q_L and L_L :

$$Q_L : 3 \text{ colors} \times 1 \text{ doublet} = 3 \text{ doublet states}, \quad (26)$$

$$L_L : 1 \text{ color singlet} \times 1 \text{ doublet} = 1 \text{ doublet state}. \quad (27)$$

Total: $3 + 1 = 4$ doublets per generation, which is even. The global $SU(2)$ anomaly is therefore evaded. $\square$

Remark 15 (HCSM framing). In HCSM terms, the number of $SU(2)_L$ doublets is the multiplicity of the B_1 isotypic content that participates in the weak interaction. The even count is a structural feature of the content as realized on the A_1 and B_1 multiplicity spaces [6].

9 Two-loop automaticity

Theorem 16 (Two-loop automatic). *By the Adler–Bardeen theorem, if the one-loop anomaly coefficients vanish, then all higher-loop anomaly coefficients vanish. In particular, the two-loop anomaly cancels automatically.*

Proof. The Adler–Bardeen theorem states that the ABJ anomaly (the anomaly of the axial current) is one-loop exact: it receives no corrections at higher loops. Consequently, the anomaly coefficient $\mathcal{A}$ satisfies $\mathcal{A} = \mathcal{A}_{\text{one-loop}}$ at all orders. Since the one-loop coefficients vanish by Theorems 2–10, the anomaly vanishes at all loops. $\square$

Remark 17 (Wess–Zumino consistency). The Wess–Zumino consistency conditions relate the anomaly coefficients across different orders. When the one-loop coefficients vanish, the Wess–Zumino conditions are trivially satisfied at all orders. The two-loop anomaly is therefore automatically zero, with no additional constraints on the content.

10 Sector-coarse framing

The anomaly conditions have a natural expression in the sector-coarse language of [6, 7].

Proposition 18 (Sector-coarse identity for the mixed color anomaly). *On the A_1 isotypic block, the mixed $\text{SU}(3)^2\text{U}(1)$ anomaly takes the sector-coarse form*

$$Y_0 + 2Y_1 + Y_2 = 0, \quad (28)$$

where Y_0, Y_1, Y_2 are the hypercharge values on the sectors $\Omega^0, \Omega^1, \Omega^2$.

Proof. The mixed $\text{SU}(3)^2\text{U}(1)$ coefficient is

$$\text{SU}(3)^2\text{U}(1) = \frac{1}{2} [n_{\text{weak}}(u_R^c) \cdot Y_0 + n_{\text{weak}}(Q_L) \cdot Y_1 + n_{\text{weak}}(d_R^c) \cdot Y_2], \quad (29)$$

where the weak multiplicities are $n_{\text{weak}}(u_R^c) = 1$ (singlet), $n_{\text{weak}}(Q_L) = 2$ (doublet), $n_{\text{weak}}(d_R^c) = 1$ (singlet). The sector assignment places u_R^c on Ω^0 , Q_L on Ω^1 , d_R^c on Ω^2 .

Substituting:

$$\text{SU}(3)^2\text{U}(1) = \frac{1}{2} [Y_0 + 2Y_1 + Y_2]. \quad (30)$$

The vanishing of this quantity is equivalent to

$$Y_0 + 2Y_1 + Y_2 = 0. \quad (31)$$

With the HCSM values $Y_0 = -2/3$, $Y_1 = +1/6$, $Y_2 = +1/3$ [7], the weighted sum is

$$-\frac{2}{3} + 2 \cdot \frac{1}{6} + \frac{1}{3} = -\frac{2}{3} + \frac{1}{3} + \frac{1}{3} = 0. \quad (32)$$

$\square$

Corollary 19 (Sector-coarse reading of anomaly cancellation). *In the sector-coarse framing, the mixed $\text{SU}(3)^2\text{U}(1)$ anomaly condition is a specific weighted sum of hypercharge values over the three Hodge sectors:*

$$Y(\Omega^0) + 2Y(\Omega^1) + Y(\Omega^2) = 0. \quad (33)$$

The weight on Ω^1 is 2, reflecting the fact that the middle sector hosts the weak doublet content (Q_L); the outer sectors Ω^0 and Ω^2 host singlet content with weight 1 each.

Remark 20 (Sector-coarse content of the other conditions). The other three anomaly conditions admit analogous sector-coarse readings:

- $\text{Tr}(Y) = 0$: the weighted sum over all states of Y vanishes.
- $\text{Tr}(Y^3) = 0$: the weighted sum over all states of Y^3 vanishes.
- $\text{SU}(2)^2\text{U}(1) = 0$: the weighted sum over the weak-doublet content vanishes.

In each case, the vanishing is due to an exact rational cancellation among the hypercharge values, reflecting the anomaly-free structure of the hypercharge functional.

11 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision, and are exact in rational arithmetic. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-08_verification.py`. It constructs the content per generation and verifies:

- (1) $\text{Tr}(Y) = 0$ (Theorem 2).
- (2) $\text{Tr}(Y^3) = 0$ (Theorem 4).
- (3) $\text{SU}(2)^2\text{U}(1) = 0$ (Theorem 7).
- (4) $\text{SU}(3)^2\text{U}(1) = 0$ (Theorem 10).
- (5) The cubic gauge anomalies vanish (Theorem 11).
- (6) The Witten global $\text{SU}(2)$ anomaly evasion (Theorem 14).
- (7) The sector-coarse identity $Y_0 + 2Y_1 + Y_2 = 0$ (Proposition 18).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

12 Conclusion

This paper has established that the HCSM hypercharge functional is anomaly-free.

The four one-loop anomaly conditions

$$\text{Tr}(Y) = 0, \quad \text{Tr}(Y^3) = 0, \quad \text{SU}(2)^2\text{U}(1) = 0, \quad \text{SU}(3)^2\text{U}(1) = 0 \quad (34)$$

vanish identically in rational arithmetic on the Standard Model content per generation, with hypercharge values given by the HCSM sector-coarse functional [7]. The three cubic gauge anomalies $\text{SU}(3)^3$, $\text{SU}(2)^3$, $\text{SU}(3)\text{--SU}(2)\text{--U}(1)$ vanish by representation-theoretic structure: $\text{SU}(3)^3$ by the vector-like color content, $\text{SU}(2)^3$ by the safety of $\text{SU}(2)$, and the mixed trilinear anomaly by the tracelessness of the non-Abelian generators. The Witten global $\text{SU}(2)$ anomaly is evaded because each generation carries 4 $\text{SU}(2)$ doublets (an even number). Two-loop cancellation is automatic by the Adler–Bardeen theorem.

In the sector-coarse framing, the mixed $\text{SU}(3)^2\text{U}(1)$ condition reduces to the weighted sector identity

$$Y(\Omega^0) + 2Y(\Omega^1) + Y(\Omega^2) = 0, \quad (35)$$

which is satisfied by the HCSM hypercharge values $(-2/3, +1/6, +1/3)$ on the A_1 block. The weight 2 on the middle sector reflects the weak-doublet content hosted there.

The anomaly cancellation is a consistency theorem of the HCSM structure. The particle content is derived in the HCSM stack; the hypercharge functional is derived; the anomaly cancellation follows. The framework's claim is: the HCSM hypercharge functional is the unique anomaly-free functional compatible with the sector-coarse structure and the Standard Model content.

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